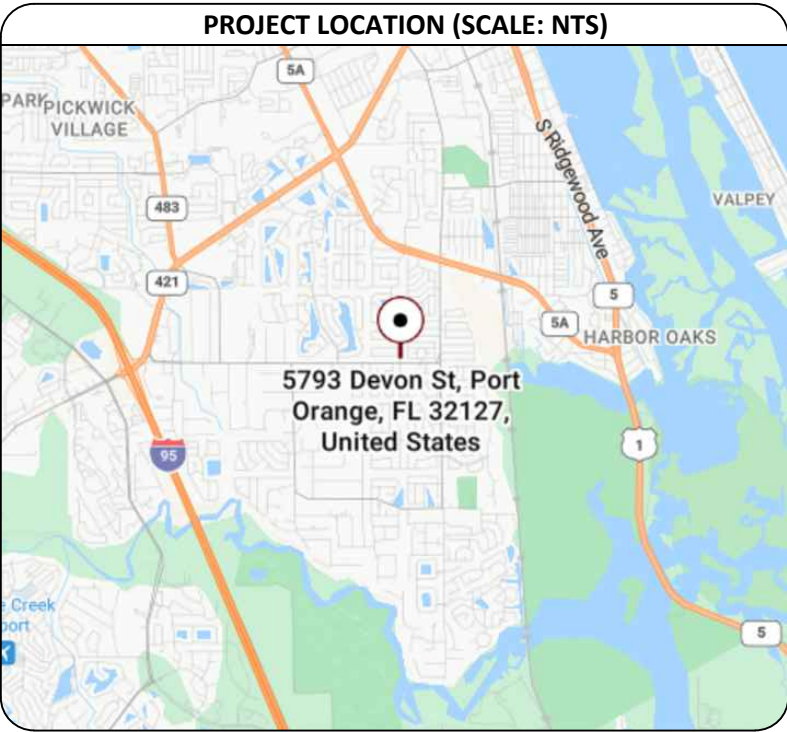


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(40) MODULES x 460W	(1) INVERTER x 16000W
18.400 kW	16.000 kW



- SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV + BATTERY SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A RESIDENTIAL LOCATION: 1234 STREET NAME, CITY, FL 9874. COORDINATES: 29.107418, -80.997862
 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM.
 - INSTALLED PV SYSTEM WILL BE CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ).
 - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

- APPLICABLE GOVERNING CODES & NOTES
- THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:
- 2023 FLORIDA BUILDING CODE (8TH EDITION)
 - 2023 FBC RESIDENTIAL (8TH EDITION)
 - 2023 FLORIDA FIRE PREVENTION CODE (8TH EDITION)
 - 2023 NATIONAL ELECTRICAL CODE (NFPA 70)
- ALL OTHER RELATED STATE & AHJ BY-LAWS & ORDINANCES

- DESIGN CRITERIA
- CONSTRUCTION: SINGLE FAMILY DWELLING (SFD)
 - ZONING: RESIDENTIAL
 - NO. OF STORY: DOUBLE STORY
 - EXPOSURE CATEGORY : B, RISK CATEGORY : II
 - WIND SPEED (ASCE 7-22) : 138MPH
 - GROUND SNOW LOAD (ASCE 7-22) : 4 lb/sqft
 - ASHRAE EXTREME LOW: -2°C
 - ASHRAE HIGH TEMP (2% AVG.) : 33°C
 - ASHRAE DISTANCE ABOVE ROOF (7/8") : 55°C

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	WIND LOAD LAYOUT
PV-4	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
S-2	EQUIPMENT ELEVATION DIAGRAM
E-1 & E-1.1	ELECTRICAL DIAGRAM & CALCULATION
E-2	PLACARDS
E-3	FIRE SAFETY PLAN
E.S	EQUIPMENT SPECIFICATIONS

EQUIPMENT DETAILS	
(#) PV MODULES	(40) APTOS SOLAR DNA-120-MF10-460W
(#) INVERTER	(1) EG4 FLEXBOSS21
(#) OPTIMIZERS	(40) TIGO (TS4-A-O) OPTIMIZERS
(#) BATTERY	(02) EG4 POWERPRO ESS BATTERIES
OUTPUT CURRENT	66.7A A
STRING	4 STRINGS OF 10 MODULES
ROOF RACKING SYSTEM	S5! S-5-PVKIT-2.0 W/ S5! S5-N
POINT OF INTERCONNECTION	FEED THROUGH LUGS IN THE EXTERIOR MAIN SERVICE PANEL WITH 200A BUSBAR, 200A MAIN BREAKER, 120/240V, 3W, 1φ
UTILITY	----
AHJ	PORT ORANGE CITY

- GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & NEC CERTIFIED, WHERE WARRANTED
 - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS.
 - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24.
 - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC
 - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM.
 - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE.
 - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.
 - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15.
 - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92.
 - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE.
 - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM.
 - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
- PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

- ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (NEC690.4(E) AND 705.6)
 - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF.
 - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES.
 - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL.
 - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS .
 - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING
 - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES.
 - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT.
 - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO NEC.
 - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED.
 - ALL EXTERIOR CONDUIT TO BE PAINT MATCHED TO THE HOMES COLORS

CONTRACTOR INFO:

DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT NAME RESIDENCE

1234 STREET NAME, CITY, FL 9874

APN: 147852369

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

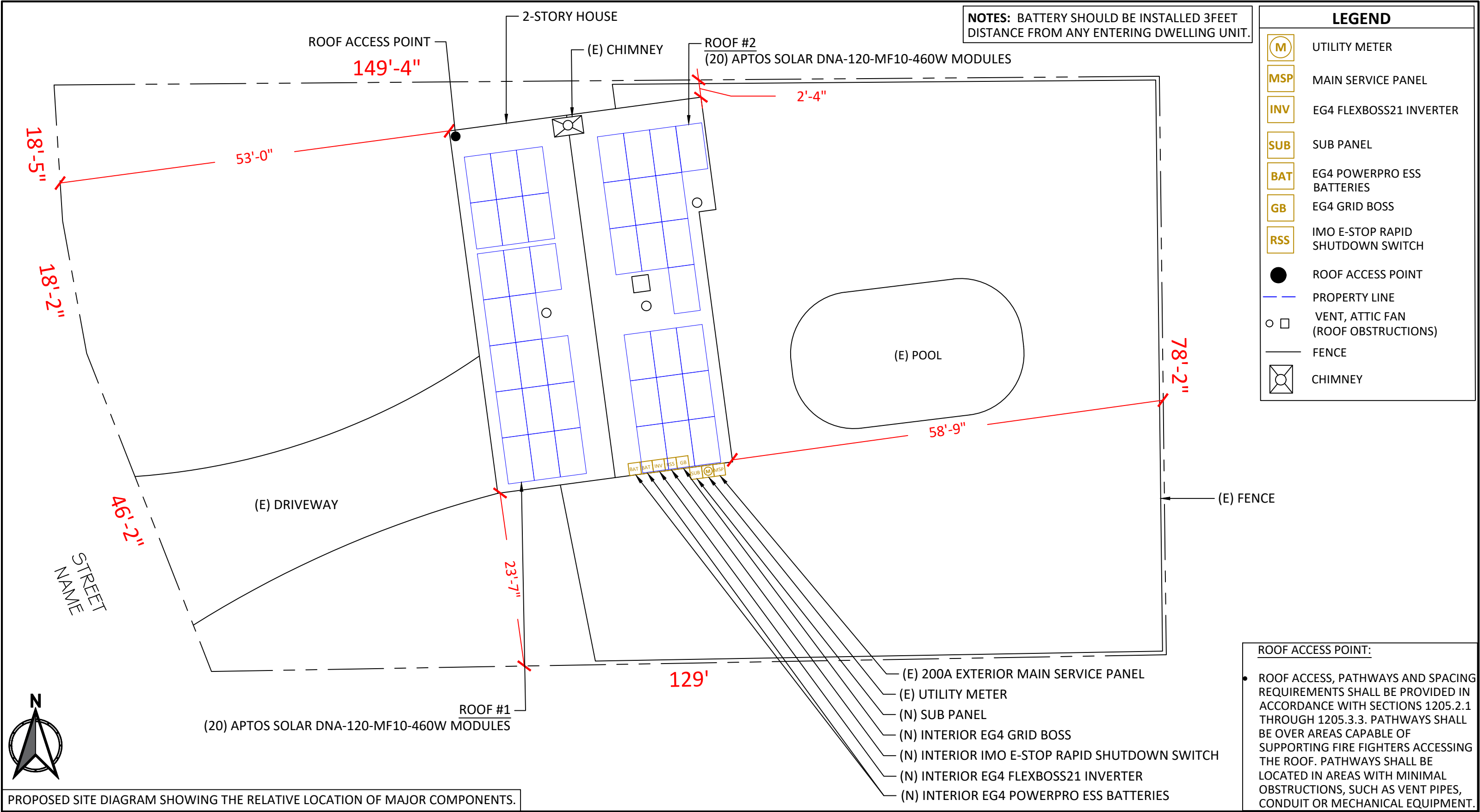
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PV-0

SHEET NO.

PROJECT SUMMARY

SHEET INFORMATION



CONTRACTOR INFO:	DATE	10/06/2025	CLIENT NAME RESIDENCE 1234 STREET NAME, CITY, FL 9874 APN: 147852369	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	PV-1 SHEET NO. SITE PLAN SHEET INFORMATION						
	PROJECT ID	SP-1		CREATED BY	KK	CHECKED BY			SS	SCALE	1/12" = 1'-0"	SHEET SIZE	11"X 17"	HOMEOWNER INFO
	CREATED BY	KK		CHECKED BY	SS	SCALE			1/12" = 1'-0"	SHEET SIZE	11"X 17"	HOMEOWNER INFO		
	CHECKED BY	SS		SCALE	1/12" = 1'-0"	SHEET SIZE			11"X 17"	HOMEOWNER INFO				
	SCALE	1/12" = 1'-0"												
SHEET SIZE	11"X 17"	HOMEOWNER INFO												

BARN ROOF DETAILS		
MOUNTING PLANE	ROOF #1	ROOF #2
NO. OF MODULES	20	20
AZIMUTH	262°	82°
ROOF TILT	18°	18°
ROOF TYPE	STANDING SEAM METAL	STANDING SEAM METAL
SEAMS SPACING	12" O.C	12" O.C
NEW ARRAY AREA	465.20	465.20
(TOTAL ARRAY AREA/ TOTAL ROOF AREA) X 100% = (930.41/1616.63) X 100% = 57.55%		

FRONT OF RESIDENCE
STREET
NAME

ROOF ARRAY
(40) APTOS SOLAR DNA-120-MF10-460W MODULES

(40) TIGO (TS4-A-O) OPTIMIZERS
(162) S5! S-5-PVKIT-2.0 ROOF ATTACHMENTS

IMO DC DISCONNECT (TYP.)
EMT CONDUIT

(E) 200A EXTERIOR
MAIN SERVICE PANEL

(E) UTILITY METER

(N) SUB PANEL

(N) INTERIOR EG4 GRID BOSS

(N) INTERIOR IMO E-STOP RAPID
SHUTDOWN SWITCH

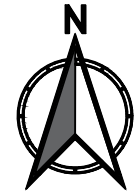
(N) INTERIOR EG4 FLEXBOSS21 INVERTER

(N) INTERIOR EG4 POWERPRO ESS BATTERIES

LEGEND	
	UTILITY METER
	MAIN SERVICE PANEL
	EG4 FLEXBOSS21 INVERTER
	SUB PANEL
	EG4 POWERPRO ESS BATTERIES BATTERY
	EG4 GRID BOSS
	IMO E-STOP RAPID SHUTDOWN SWITCH
	JUNCTION BOX
	TIGO (TS4-A-O) OPTIMIZERS
	S5! S-5-PVKIT-2.0 ATTACHMENT
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	SEAMS
	EMT CONDUIT
	CHIMNEY

NOTES: ATTACHMENT POINTS ARE NOT EXACT. CONTRACTOR MAY NEED TO ADJUST MOUNT LOCATION. EXISTING ROOF CONDITION & RAFTER LOCATIONS ARE SUBJECT TO FIELD VERIFICATION. ACTUAL LOCATION MAY DIFFER. INSTALL PER MANUFACTURER SPECIFICATIONS. SEE SUPPLEMENTAL DOCUMENTS FOR ATTACHMENT & STRUCTURAL DOCUMENTATION.

NOTES: ROOF ACCESS, PATHWAYS AND SPACING REQUIREMENTS SHALL BE PROVIDED IN ACCORDANCE WITH SECTIONS 1205.2.1 THROUGH 1205.3.3. PATHWAYS SHALL BE OVER AREAS CAPABLE OF SUPPORTING FIRE FIGHTERS ACCESSING THE ROOF. PATHWAYS SHALL BE LOCATED IN AREAS WITH MINIMAL OBSTRUCTIONS, SUCH AS VENT PIPES, CONDUIT OR MECHANICAL EQUIPMENT.



NOTE:
MINIMUM 3' CLEARANCE BETWEEN BATTERY AND DOOR/WINDOW DIRECTLY ENTERING THE DWELLING UNIT.

CONTRACTOR INFO:	DATE	10/06/2025
	PROJECT ID	SP-1
	CREATED BY	KK
	CHECKED BY	SS
	SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"	
	ANSI B	

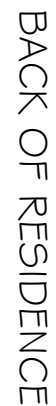
DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

CLIENT NAME RESIDENCE		
1234 STREET NAME, CITY, FL 9874		
APN: 147852369		
HOMEOWNER INFO		

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY	

PV-2
SHEET NO.
ROOF ATTACHMENT DETAILS
SHEET INFORMATION



PV-3

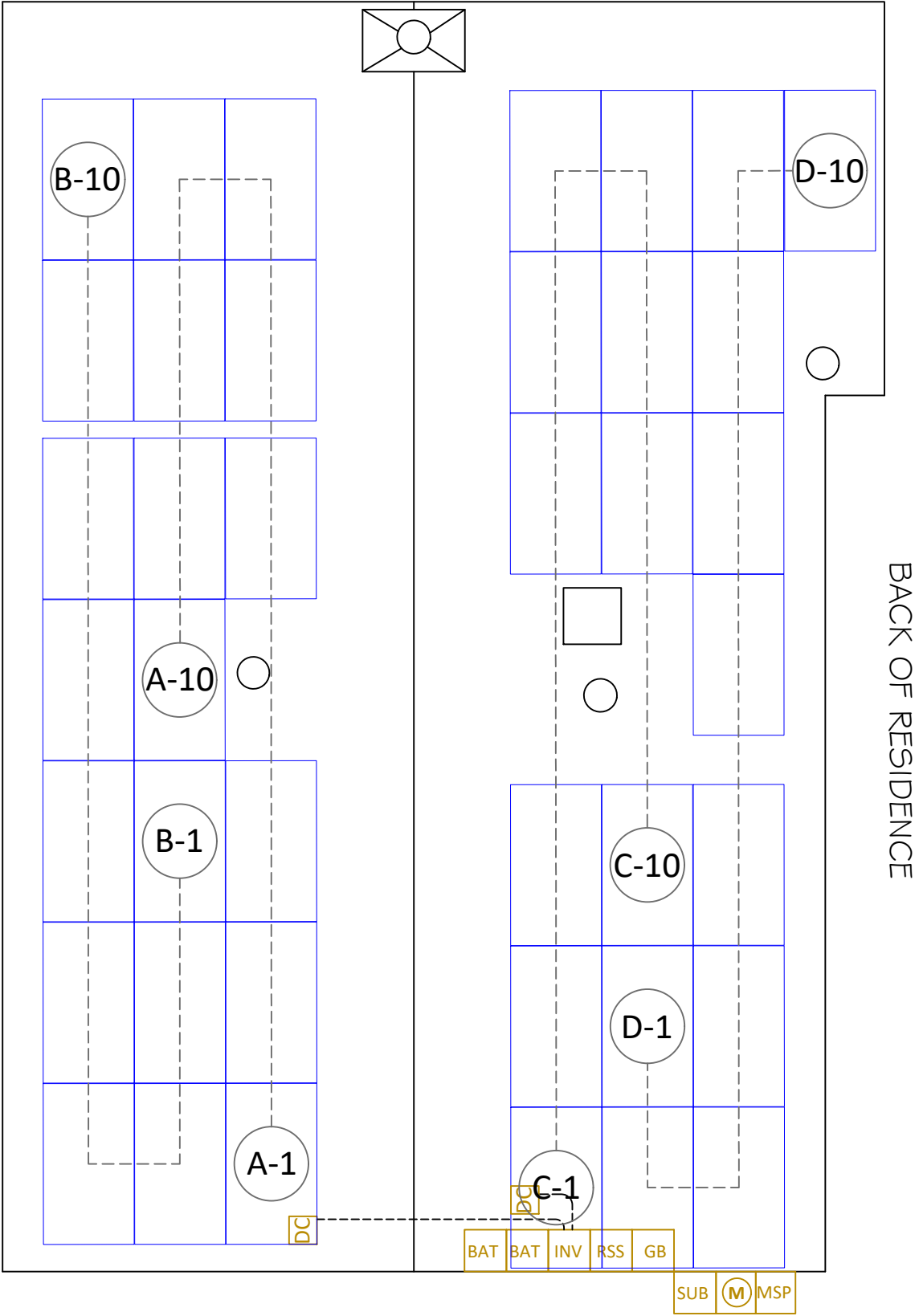
WIND LOAD LAYOUT

BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
SOLAR PV MODULE	40	APTOS SOLAR DNA-120-MF10-460W MODULES
OPTIMIZERS	40	TIGO (TS4-A-O) OPTIMIZERS
INVERTER	1	EG4 FLEXBOSS21
BATTERY	2	EG4 POWERPRO ESS BATTERIES
IMO DC DICONNECT	2	IMO DC DICONNECT ROOFTOP ISOLATOR SWITCH
RAPID SHUTDOWN SWITCH	1	IMO E-STOP RAPID SHUTDOWN SWITCH
SUB PANEL	1	200A SUB PANEL, 240V, NEMA 3R, UL LISTED
GATEWAY	1	EG4 GRIDBOSS MID V3.1
ATTACHMENT	162	S5! S5-N
ATTACHMENT	162	S5! S-5-PVKIT-2.0-EDGE KIT
ATTACHMENT	162	S5! S-5-PV KIT-2.0-MID BLACK KIT
GROUNDING LUG	10	GROUNDING LUG

ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	40	
Module Weight	52.9	LBS
Total Module (Array) Weight	2116.00	LBS
Number of Attachment point	162	
Mounting System Weight (Per Module)	1.5	LBS
Mounting System Weight	243.00	LBS
Total System Weight (Module Weight + Mounting System Weight)	2359.00	LBS
Weight at Each Attachment Point (Array Weight / Number of Attachment Point)	13.06	LBS
Module Area (75.1"x44.6")	23.26	SqFt
Total Array Area	930.41	SqFt
Distributed Load (Total System Weight / Total Array Area)	2.54	Per SqFt
New pv array area	930.41	SqFt
Total Roof Area	1616.63	SqFt
Total Percentage of Roof Covered Total Array Area / Total Roof Area)*100	57.55%	



FRONT OF RESIDENCE
STREET NAME



LEGEND	
	UTILITY METER
	MAIN SERVICE PANEL
	EG4 POWERPRO ESS BATTERIES
	EG4 FLEXBOSS21
	SUB PANEL
	IMO DC DISCONNECT
	EG4 GRID BOSS
	RAPID SHUTDOWN SWITCH
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	EMT CONDUIT
	CHIMNEY

LEGEND	
	CIRCUIT #1 - 10 MODULES
	CIRCUIT #2 - 10 MODULES
	CIRCUIT #3 - 10 MODULES
	CIRCUIT #4 - 10 MODULES

NOTE:
MINIMUM 3' CLEARANCE BETWEEN BATTERY AND DOOR/WINDOW DIRECTLY ENTERING THE DWELLING UNIT.

		DATE	10/06/2025
		PROJECT ID	SP-1
		CREATED BY	KK
		CHECKED BY	SS
		SCALE	1/10" = 1'-0"
CONTRACTOR INFO:		SHEET SIZE	11"X 17"
			ANSI B

DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	1/10" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

		CLIENT NAME RESIDENCE	
		1234 STREET NAME, CITY, FL 9874	
		APN: 147852369	
		HOMEOWNER INFO	

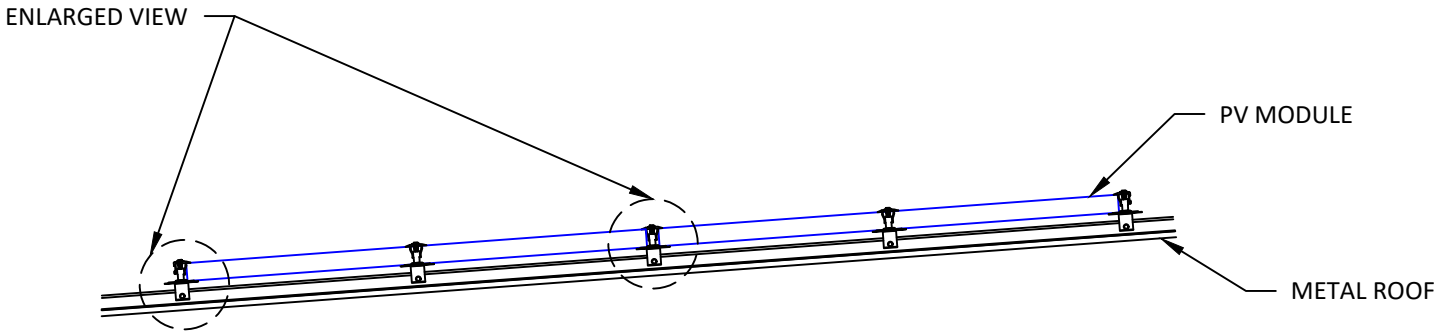
#	REVISION/UPDATE	DATE

		SIGNED & STAMP BY

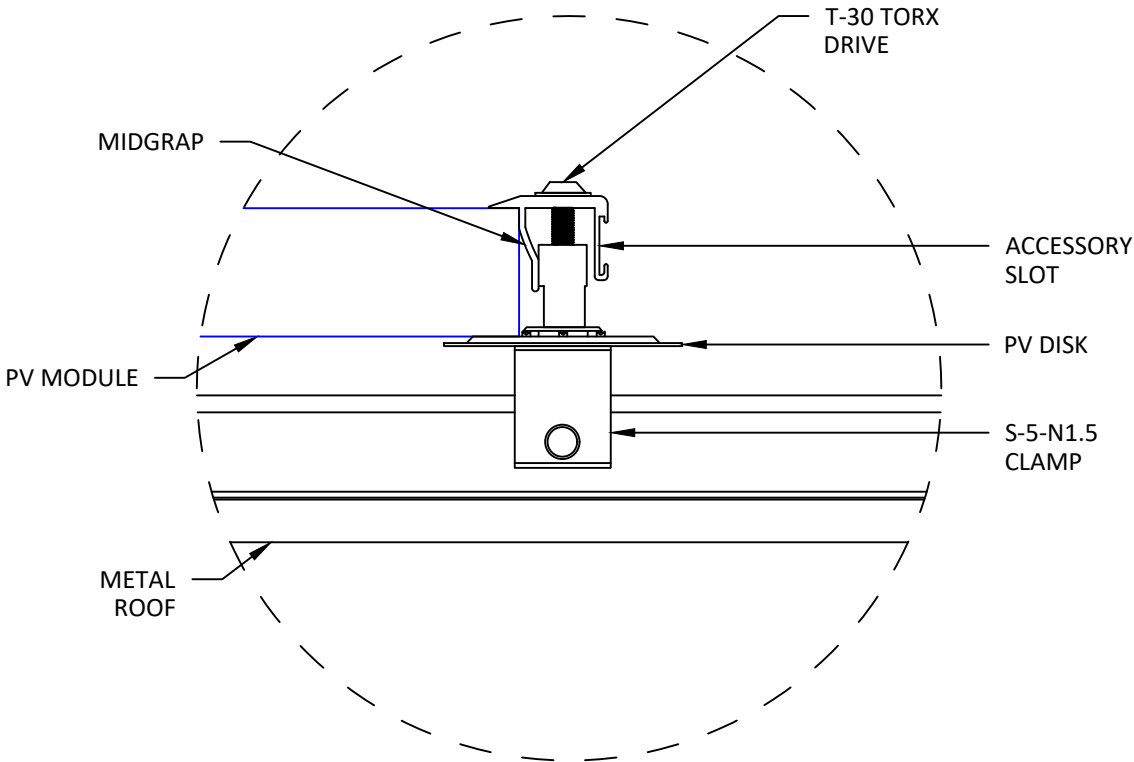
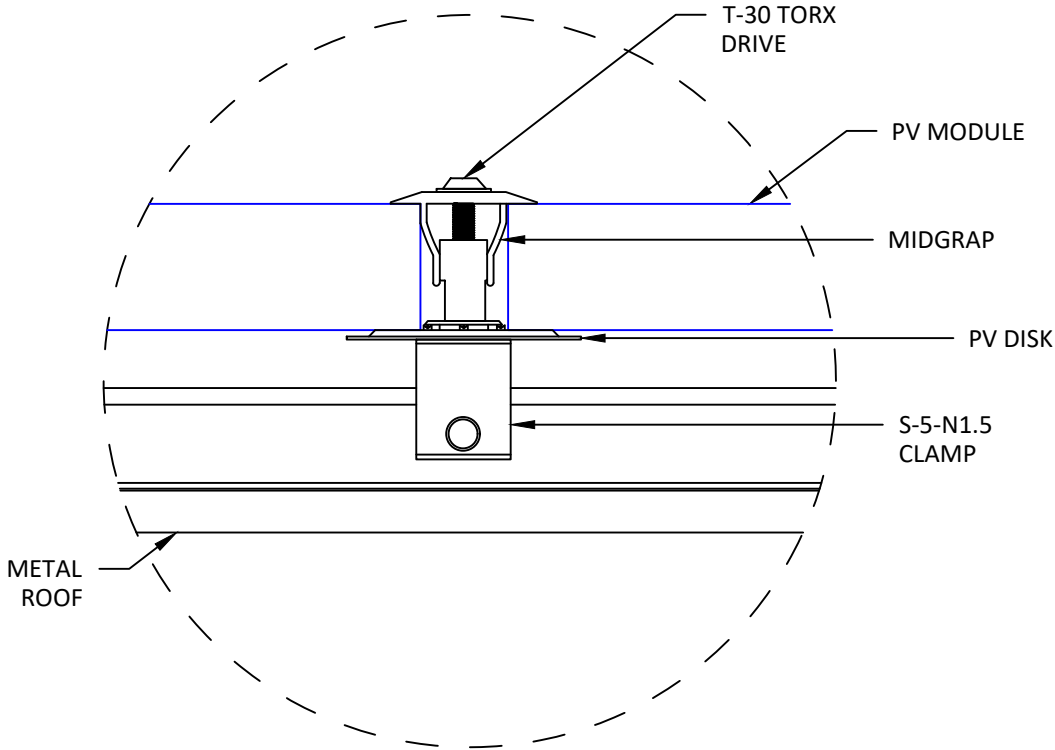
		PV-4

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE SYSTEM S5! S-5-PVKIT-2.0 ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. THE PROPOSED PV SYSTEM ADDS 2.49 PSF TO THE ROOF FRAMING SYSTEM.
- 4. ROOF LIVE LOAD = 20 PSF
- 5. GROUND SNOW LOAD = 4 lb/sqft
- 6. WIND SPEED = 138 MPH
- 7. EXPOSURE CATEGORY =B
- 8. RISK CATEGORY = II



STRUCTURAL ELEVATION DIAGRAM



ATTACHMENT DETAIL (SIDE VIEW)

NOTES: FLASHING SHALL BE APPLIED IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.
IF THERE IS ANY CONFLICT BETWEEN WHAT IS DEPICTED & INSTRUCTION AS PER MANUFACTURER, MANUFACTURERS INSTRUCTIONS SHALL SUPERSEDE.

CONTRACTOR INFO:

DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT NAME RESIDENCE
1234 STREET NAME, CITY, FL 9874
APN: 147852369

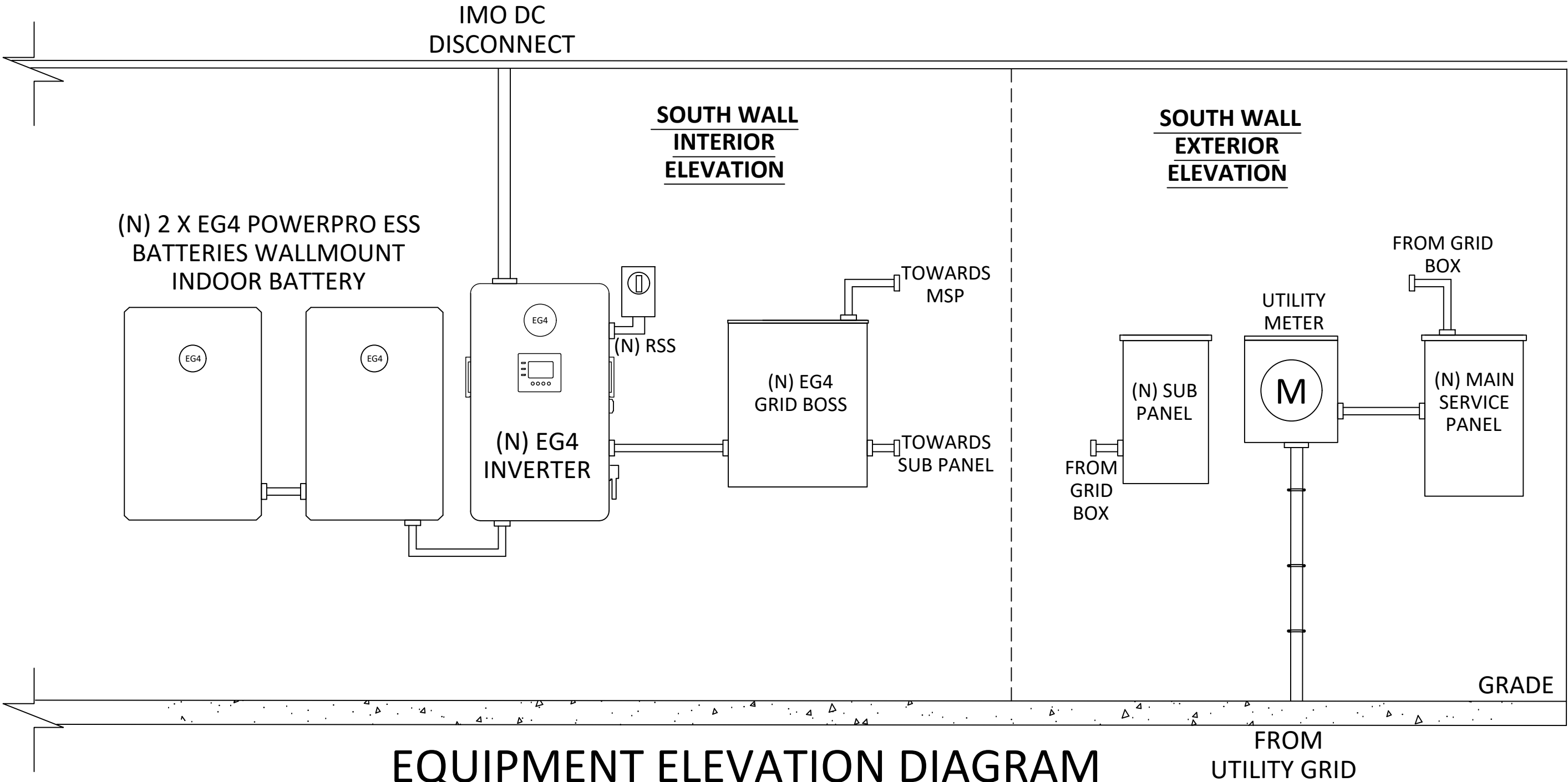
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-1
SHEET NO.
STRUCTURAL
DETAILED
DIAGRAM
SHEET INFORMATION

NOTES:
ESS UNIT LOCATION(S) AND DISTANCES RELATIVE TO ONE ANOTHER AND TO ALL RELEVANT BUILDING FEATURES AND OTHER EQUIPMENT, INCLUDING BUT NOT LIMITED TO THE FOLLOWING:
I.) INDIVIDUAL ESS UNITS SHALL BE SEPARATED FROM ONE ANOTHER BY 6.69".
II.) WHERE AN ESS IS INSTALLED ON AN EXTERIOR WALL OR OUTDOOR, ESS UNITS SHALL BE SEPARATED FROM ALL WINDOWS, DOORS, OPERABLE OPENINGS AND HVAC INLETS BY 3 FEET (IN ANY DIRECTION).

NOTES: INDIVIDUAL ESS UNITS SHALL BE SEPARATED FROM EACH OTHER BY A MIN. OF 6.69" OR PER FIRE DEPT. APPROVED PLANS.



CONTRACTOR INFO:	DATE	10/06/2025
	PROJECT ID	SP-1
	CREATED BY	KK
	CHECKED BY	SS
	SCALE	NTS
	SHEET SIZE	11"X 17"

SHEET SIZE	ANSI B
------------	--------

HOMEOWNER INFO
CLIENT NAME RESIDENCE
1234 STREET NAME, CITY, FL 9874
APN: 147852369

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-2
SHEET NO.
EQUIPMENT ELEVATION DIAGRAM
SHEET INFORMATION

PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(40) MODULES x 460W	(1) INVERTER x 16000W
18.400 kW	16.000 kW
BATTERY SPECIFICATION	
MODEL	EG4 POWERPRO ESS BATTERIES
USABLE CAPACITY	14.3 KWH
INTERCONNECTION	240VAC

INVERTER SPECIFICATIONS		
MANUFACTURER / MODEL	EG4 FLEXBOSS21	
NOMINAL OUTPUT VOLTAGE	240VAC	
CONTINUOUS OUTPUT CURRENT	66.7A A	

AMBIENT TEMPERATURE SPECS		
RECORD LOW TEMP	-2°C	
AMBIENT TEMP (HIGH TEMP 2%)	33°C	
CONDUIT HEIGHT	7/8"	
ROOF TOP TEMP	90°C	
CONDUCTOR TEMPERATURE RATE	55°C	
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.27%/°C	

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	APTOS SOLAR DNA-120-MF10-460W
VMP	35.36V
IMP	13.00
VOC	41.97V
ISC	13.75A
TEMP. COEFF. VOC	-0.27%/°C
MODULE DIMENSION	75.1" (L) x 44.6" (W)
PANEL WATTAGE	460W

EQUIPMENT DETAILS	
(#) PV MODULES	(40) APTOS SOLAR DNA-120-MF10-460W
(#) OPTIMIZERS	(40) TIGO (TS4-A-O) OPTIMIZERS
(#) INVERTER	(1) EG4 FLEXBOSS21
(#) BATTERY	(2) EG4 POWERPRO ESS BATTERIES
OUTPUT CURRENT	66.7A A
STRING	4 STRINGS OF 10 MODULES
POINT OF INTERCONNECTION	FEED THROUGH LUGS IN THE EXTERIOR MAIN SERVICE PANEL WITH 200A BUSBAR, 200A MAIN BREAKER, 120/240V, 3W, 1φ
UTILITY	----
AHJ	PORT ORANGE CITY

THE SUMMATION RULE - NEC - 705.12 (B)(3)(3)

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT x 1.25				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	PV WIRE		OPEN AIR	1	2	N/A	N/A	6 AWG	THWN-2, COPPER	0.76	(55°C)	N/A	13.42	x	1.25	=	16.7A	40A	N/A	70FT	1.02%
B	10 AWG	THWN-2	COPPER	MIN 0.75" Dia PVC	2	4	25.63%	N/A	6 AWG	THWN-2, COPPER	0.76	(55°C)	0.8	13.42	x	1.25	=	16.7A	40A	40A x 0.8 x 0.76 = 24.3A	24FT	0.35%
C	3 AWG	THWN	COPPER	MIN 1" Dia PVC	1	3	39.77%	90A	6 AWG	THWN, COPPER	0.96	(33°C)	1	66.7A	x	1.25	=	83.4A	115A	115A x 1 x 0.96 = 110.4A	5FT	0.08%
D	4/0 AWG	THWN	COPPER	MIN 2" Dia PVC	1	3	31.29%	200A	2 AWG	THWN, COPPER	0.96	(33°C)	1	160.0A	x	1.25	=	200.0A	260A	260A x 1 x 0.96 = 249.6A	5FT	0.06%
E	4/0 AWG	THWN	COPPER	MIN 2" Dia PVC	1	3	31.29%	N/A	2 AWG	THWN, COPPER	0.96	(33°C)	1	160.0A	x	1.25	=	200.0A	260A	260A x 1 x 0.96 = 249.6A	5FT	0.06%

CONTRACTOR INFO:	DATE	10/06/2025	CLIENT NAME RESIDENCE 1234 STREET NAME, CITY, FL 9874 APN: 147852369 HOMEOWNER INFO	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-1.1 SHEET NO. ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION
	PROJECT ID	SP-1						
	CREATED BY	KK						
	CHECKED BY	SS						
	SCALE	NTS						
	SHEET SIZE	11"X 17"						
		ANSI B						

1

WARNING

ELECTRIC SHOCK HAZARD

DO NOT TOUCH TERMINALS

TREMINALS ON BOTH THE LINE AND LOAD SIDE MAY BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
COMBINER BOX/ EMT ENCLOSURES/
AC DISCONNECT/ MAIN SERVICE PANEL
(PER CODE: NEC 2023, 690.13(B))

2

WARNING

PHOTOVOLTAIC POWER SOURCE

LABEL LOCATION:
CONDUIT, RACEWAY, ENCLOSURES,
COMBINER BOX & AC DISCONNECT
(PER CODE: NEC 2023, 690.31(G)(3)(4))

3

PHOTOVOLTAIC

AC DISCONNECT

LABEL LOCATION:
AC DISCONNECT/ BREAKER/
POINTS OF CONNECTION
(PER CODE: NEC 2023, 690.13(B))

4

PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT66.7 A

NOMINAL OPERATING AC VOLTAGE240 VAC

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT
2023 NEC 690. 13(B) & 690.54

5

RAPID SHUTDOWN SWITCH
FOR SOLAR PV SYSTEM

LABEL LOCATION:
RAPID SHUTDOWN (AC DISCONNECT)
PER CODE: NEC 690.56 (C)(3)

6

WARNING

DUAL POWER SUPPLY

SOURCES: UTILITY GRID
AND PV POWER SOURCE

ELECTRIC SYSTEM

LABEL LOCATION:
POINT OF INTERCONNECTION
(PER CODE: NEC 2023, 705.12(B)(3))

7

EMERGENCY RESPONDER

SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM

SOLAR ELECTRIC
PV PANELS

LABEL LOCATION:
RAPID SHUTDOWN (AC DISCONNECT)
PER CODE: NEC 690.56 (C)(1) AND
NFPA 11.1.12.2.1.1.1.1 & 11.12.2.1.4

8

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES.
TOTAL RATING OF ALL OVERCURRENT DEVICES,
EXCLUDING MAIN SUPPLY OVERCURRENT DEVICES
SHALL NOT EXCEED AMPACITY OF BUSBAR.

REQ'D BY: NEC 705.12(B)(3)(3)

9

CAUTION - TRI POWER SOURCE

FIRST SOURCE IS UTILITY ELECTRICAL GRID
SECOND SOURCE IS AC BATTERY
THIRD SOURCE IS PV INVERTER

LABEL LOCATION:
SUB PANEL OR BACKUP LOADS PANEL, BATTERY, INVERTER OUTPUT
PANEL, MAIN SERVICE PANEL, METER/MAIN AND PV AC DISCONNECT
2023 NEC 705.10

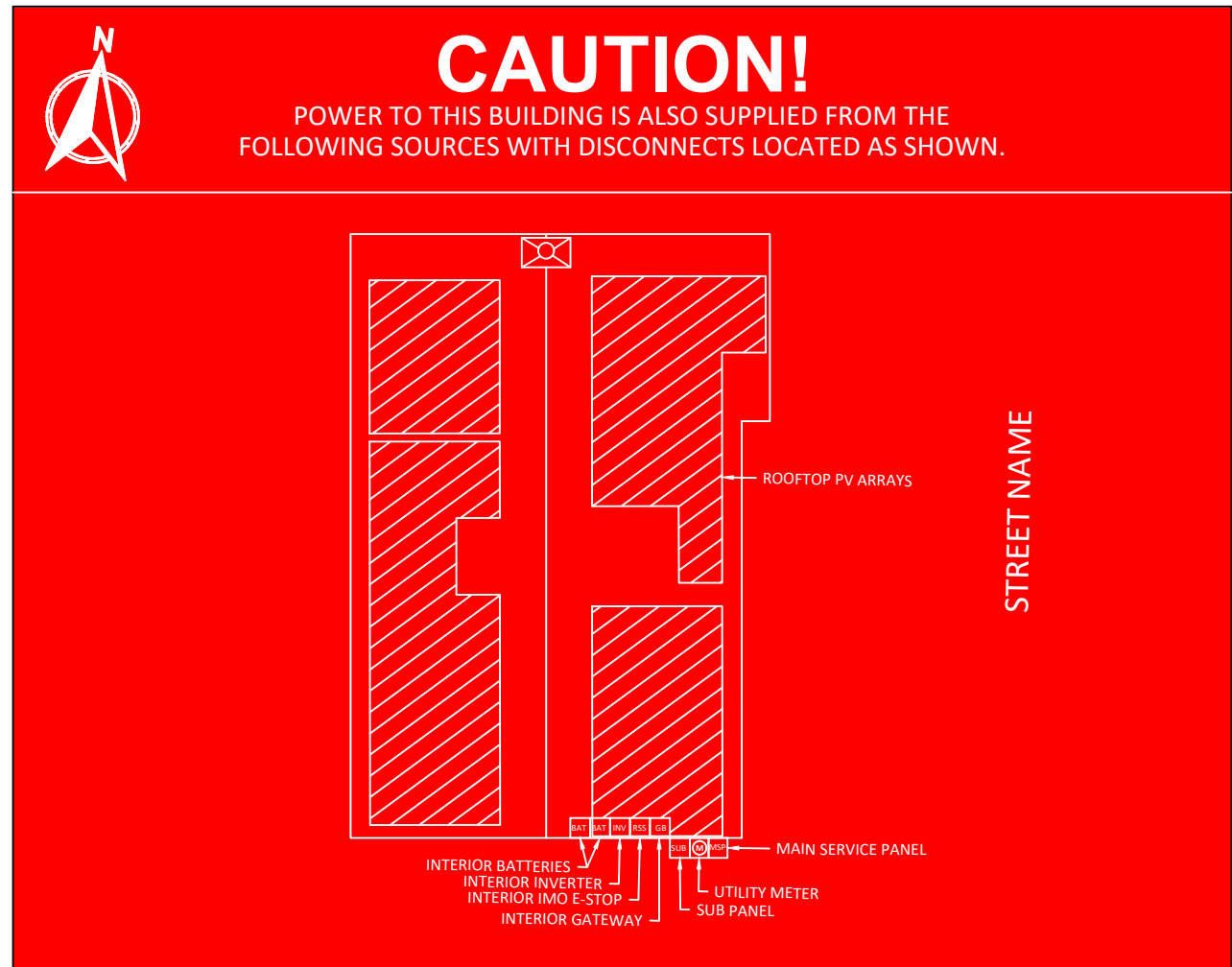
10

BATTERY DISCONNECT

DO NOT RELOCATE
THIS OVERCURRENT
DEVICE

REQ'D BY: NEC 705.12(B)(2)(3)(b)
LABEL LOCATION:
BATTERY BREAKERS

- ADHESIVE FASTENED SIGNS:
- ANSI Z535.4-2011 PRODUCT SAFETY SIGNS AND LABELS, PROVIDES GUIDELINES FOR SUITABLE FONT SIZES, WORDS, COLORS, SYMBOLS, AND LOCATION REQUIREMENTS FOR LABELS. NEC 110.21(B)(1)
 - THE LABEL SHALL BE OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED. NEC 110.21(B)(3)
 - ADHESIVE FASTENED SIGNS MAY BE ACCEPTABLE IF PROPERLY ADHERED. VINYL SIGNS SHALL BE WEATHER RESISTANT.



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

- ADHESIVE FASTENED SIGNS:
- ANSI Z535.4-2011 PRODUCT SAFETY SIGNS AND LABELS, PROVIDES GUIDELINES FOR SUITABLE FONT SIZES, WORDS, COLORS, SYMBOLS, AND LOCATION REQUIREMENTS FOR LABELS. NEC 110.21(B)(1)
 - THE LABEL SHALL BE OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED. CEC 110.21(B)(3)
 - ADHESIVE FASTENED SIGNS MAY BE ACCEPTABLE IF PROPERLY ADHERED. VINYL SIGNS SHALL BE WEATHER RESISTANT.

CONTRACTOR INFO:

DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT NAME RESIDENCE
1234 STREET NAME, CITY, FL 9874
APN: 147852369

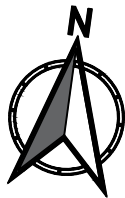
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

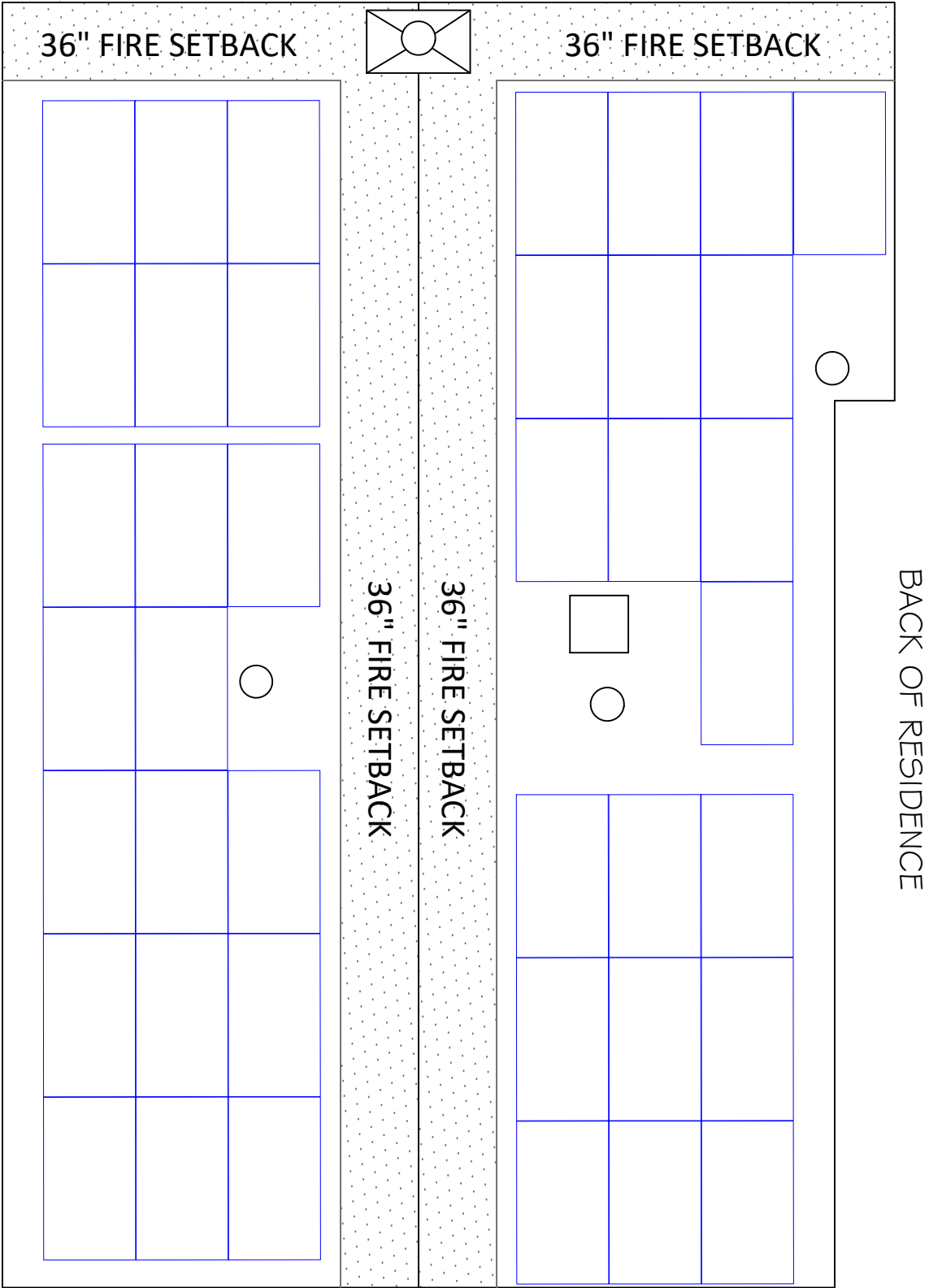
E-2
SHEET NO.

PLACARDS

SHEET INFORMATION



FRONT OF RESIDENCE
STREET NAME



GENERAL NOTES:

- EQUIPMENT LIKELY TO BE WORKED UPON WHILE ENERGIZED SHALL BE INSTALLED IN LOCATIONS THAT SATISFY MIN. WORKING CLEARANCES PER NEC 110.26
- 24/7 UNESCORTED KEYLESS ACCESS SHALL BE PROVIDED TO ALL TAMPA ELECTRIC CO EQUIPMENT.
- CONTRACTOR SHALL USE ONLY COMPONENTS LISTED BY A NATIONALLY RECOGNIZED TESTING LABORATORY FOR THE INTENDED USE.
- CONTRACTOR IS RESPONSIBLE FOR FURNISHING ALL EQUIPMENT, CABLES, ADDITIONAL CONDUITS, RACEWAYS, AND OTHER ACCESSORIES NECESSARY FOR A COMPLETE AND OPERATIONAL PV SYSTEM.
- ALL EXPOSED PV ROOFTOP CONDUCTORS NOT UNDER THE ARRAY SHALL BE PROTECTED BY A RACEWAY WITH A LISTED JUNCTION BOX AT BOTH ENDS AND COMPLY WITH NEC 690.31(A).
- ALL EMT CONDUIT FITTINGS SHALL BE LISTED AS WEATHERPROOF FITTINGS AND INSTALLED TO ENSURE A RAINTIGHT FIT, PER NEC 358.42.

CONTRACTOR INFO:

DATE	10/06/2025
PROJECT ID	SP-1
CREATED BY	KK
CHECKED BY	SS
SCALE	1/10" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT NAME RESIDENCE
1234 STREET NAME, CITY, FL 9874
APN: 147852369

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-3
SHEET NO.
FIRE SAFETY
PLAN

SHEET INFORMATION



DNA™ 120-Monofacial

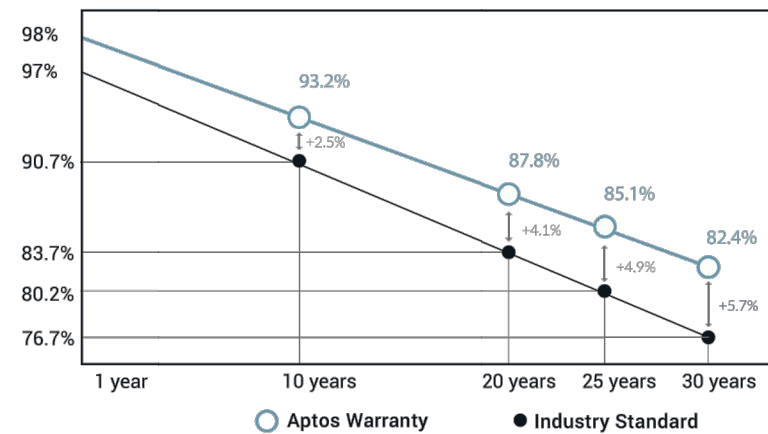
Solar for Innovators

DNA-120-MF10-460W

Residential | Commercial

Our DNA Split Cell Series uses advanced selective emitter PERC technology with thin film layers to improve heat tolerance, maximize energy harvest, minimize resistive loss, and use 5% more of the available active area for optimal power performance.

Linear Performance Warranty



Key Features



Advanced Technology

Patented DNA™ technology boosts power performance & module efficiency.



Miami-Dade Approved

Maximum Durability: LEVEL 6 SALT MIST Tested, 5400 Pa Wind Load Certified Panels



Bankable Investment

Comprehensive warranty that covers both **30-year product** and **30-year power** performance.



Awards

Winners of the Leadership in Solar Energy award for three consecutive years and listed as one of the Top Solar Products from 2021-2022.



Aesthetics

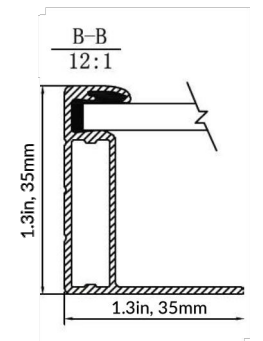
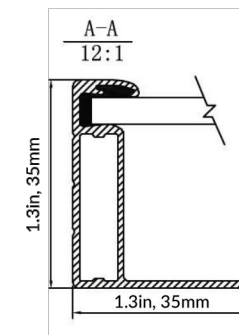
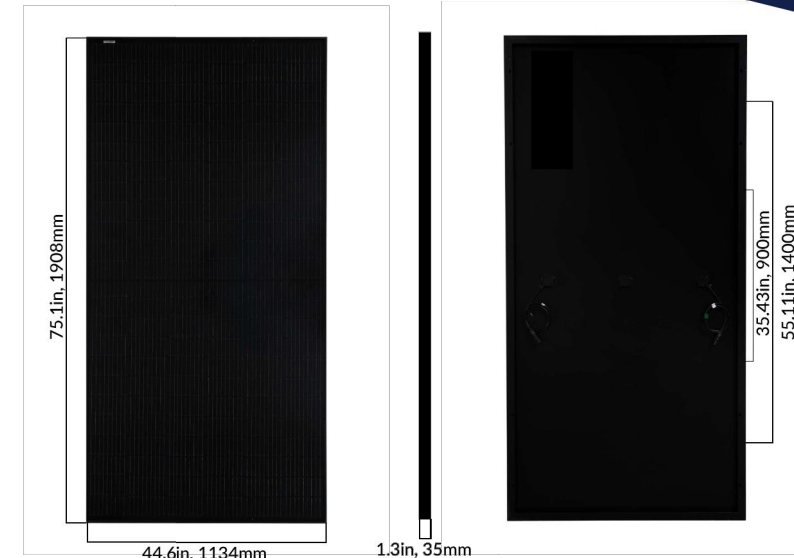
All black design with advanced split cell technology features 10 ultra-thin busbars that allow for less resistance and greater energy harvest.

450W | 455W | 460W



Designed & Engineered In Silicon Valley

DNA™ 120 Monofacial



Electrical Specifications	DNA-120-MF10-450W		DNA-120-MF10-455W		DNA-120-MF10-460W	
	STC	NOCT	STC	NOCT	STC	NOCT
STC Rated Output P _{mpp} (W)	450W	336W	455W	341W	460W	344W
Open Circuit Voltage V _{voc} (V)	41.79	39.50	41.88	39.58	41.97	39.67
Short Circuit Current I _{sc} (A)	13.51	11.99	13.63	12.09	13.75	12.20
Rated Voltage V _{mp} (V)	35.15	32.39	35.27	32.46	35.36	32.53
Rated Voltage I _{mp} (A)	12.80	10.40	12.90	10.49	13.00	10.58
Module Efficiency	20.79%		21.03%		21.26%	
STC for front-face of panel: 1000 W/m², 25°C, measurement uncertainty ±3% NOCT for front-face of panel: 800W/m², 45°C, Wind speed 1 m/s						

Temperature Coefficients

Temperature Coefficients P _{mpp}	-0.35%/°C
Temperature Coefficients I _{sc}	+0.054%/°C
Temperature Coefficients V _{oc}	-0.27%/°C
Nominal Operating Cell Temperature (NOCT)	45±2°C

Test Operating Conditions

Maximum Series Fuse	25A
Maximum System Voltage	1,500 VDC (UL&IEC)
Maximum Load Capacity (Per UL 1703)	5400 PA Snow Load / 5400 Pa Wind Load
Fire Performance Type	Type 1

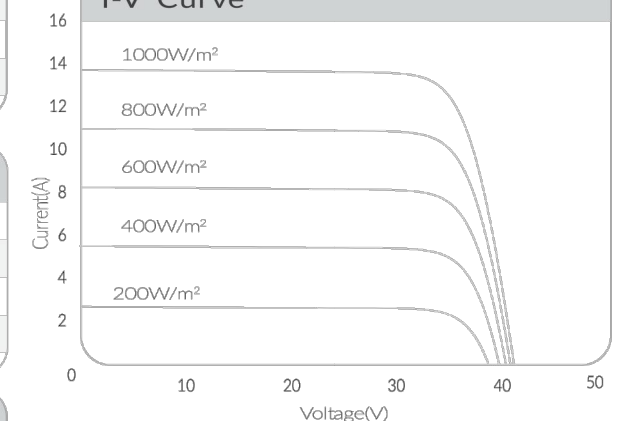
Packaging Configuration

Number of Modules per Pallet	31
Number of Pallets per 40ft. Container	24
Pallet Dimensions	76.29 X 44.48 X 49.76 in, 1938 X 1130 X 1264 mm
Pallet Weight (lbs)	1640
Module per 40ft Container	744

Mechanical Properties

Cell Type	SE-PERC
Glass	3.2mm, anti-reflection coating, high transmission, low iron, tempered glass
Frame	Anodized Aluminum Alloy
Junction Box	IP68
Dimensions	75.1 X 44.6 X 1.3 in, 1908 X 1134 X 35 mm
Weight	52.9lbs.(24.2kg)
Output Cable	4mm2 (EU)12AWG,39.37in.(1200mm)
Cable Length	47.2 in
Encapsulant	POE
Connector Type	Staubli EVO2

I-V Curve



Certifications



8207 Callaghan Rd, Ste 100,
San Antonio, Texas 78230
www.aptossolar.com | sales@aptossolar.com

Aptos Solar Technology reserves the right to make specification changes without notice.

Ref_11.13.24



TS4-A-O

Module-level PV Optimizer

The TS4-A-O (Optimization) is the advanced add-on optimization solution that brings smart module functionality to standard PV modules for higher reliability. Improve energy efficiency by upgrading underperforming PV systems or adding smart features to new installations.

The TS4-A-O complies with NEC 2017 and 2020 690.12 Rapid Shutdown specifications when installed on each module and accompanied by the Tigo Access Point (TAP) and the Cloud Connect Advanced (CCA). Module-level DC production data, as well as data from other Modbus connected devices, can be analyzed via Tigo's Energy Intelligence Software when connected to the cloud.

The TS4-A-O with UHD-Core technology and expanded specifications supports PV modules up to 700W, and 15A.

Included Features



Module-level **optimization** for increased energy yield and greater design flexibility



Manual or automatic module-level **shutdown**. Complies with NEC 2017 and 2020.



Module-level **monitoring** for energy production tracking and system management

Easy Installation

Snap to standard module frame or remove clips for rack mounting

Smart Commissioning

Configure and commission with your Android or iOS mobile device



TS4-A-O SPECIFICATIONS

Environmental

Operating Temperature Range -40°C to +70°C (-40°F to +158°F)

Outdoor Rating IP68, NEMA 3R

Maximum Elevation 2000m

Mechanical

Dimensions W=138.4mm, L= 139.7mm, H= 22.9mm

Weight 520g

Electrical

Max Input Voltage (V_{oc} @ Lowest Temperature) 80V

Input Voltage Range 16 - 80V*

Maximum Cont. Current (I_{max}) 15A

Maximum Short Circuit Current (I_{sc}) 20A

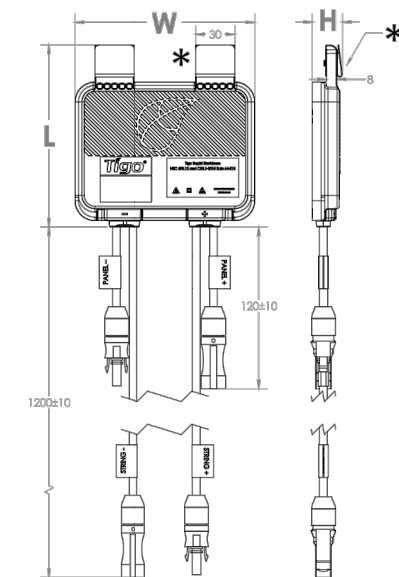
Maximum Power 700W

Cable Length (in/out) 0.12/1.2m (standard), 0.62/1.2m (optional)

Connectors MC4 (standard), EVO2 (optional)

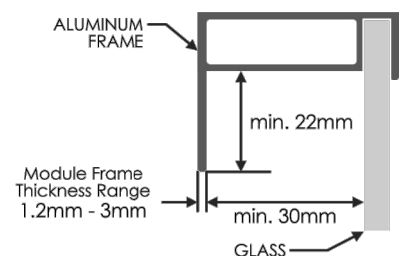
Communication Type Wireless

Recommended Fuse Rating 30A



*Clips can be removed for rack mounting

Module frame specifications for mounting TS4-A



TAP required for module-level shutdown and CCA required for monitoring with TS4-A-O.

*V_{MOD MAX} of the connected panel = TS4-A-O V_{DCU MAX}

**Limits are based on NEC 690.12 rapid shutdown requirements.

ORDERING INFORMATION

PVC00101-A 1500V TÜV, 1.2m Kabel, EVO2



For sales info:

sales@tigoenergy.com

For product info:

Visit tigoenergy.com/products

For technical info:

Visit support.tigoenergy.com

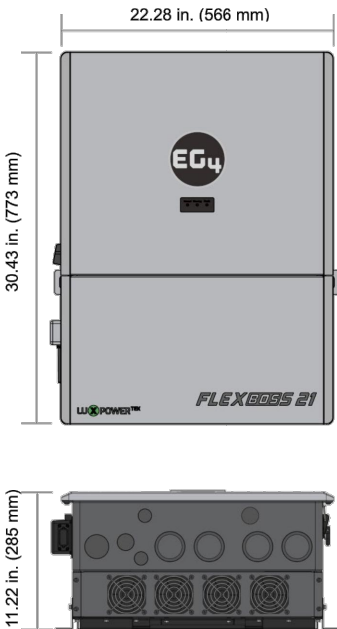
For additional info and product selection assistance, use Tigo's online design tool at tigoenergy.com/design



PV 2.0

Tigo[®]

Tigo Energy, Inc. | www.tigoenergy.com | sales@tigoenergy.com



EG4® FLEXBOSS21 HYBRID INVERTER

The EG4 FlexBOSS21 is a versatile 48V split-phase, hybrid inverter/charger that offers the same dependable power as the 18kPV with enhanced flexibility. Powerful enough to start a 5-ton AC unit, the FlexBOSS21 supports up to 21kW of PV input. Capable of paralleling up to 16 units together, the FlexBOSS21 has an impressive total output of 256kW. Able to provide 16kW of continuous output power with PV & battery, and up to 12kW continuous output by using battery alone. Three individual MPPTs give users optimal control over their solar needs, while the updated EG4 monitoring software allows for convenient total remote management, complete with mobile notifications and remote setting. Seamless interaction with the EG4 GridBOSS gives users control over the entire Energy Storage System (ESS).

HIGH
FREQUENCY
SPLIT-PHASE
DESIGN

* 10-YEAR
WARRANTY

REMOTE
ADJUSTMENT
VIA EG4
SOFTWARE

ALL-IN-ONE HYBRID INVERTER

Capable of running entirely off grid, using grid electricity, and selling power back to the grid.

UP TO 600VDC INPUT

The extra high voltage enables lower cable sizing for the 3 MPPTs and a maximum recommended PV input of 21kW, eliminating the need for a combiner box.

PLUG-IN WI-FI DEVICE

Enables wireless connection between our monitoring platform and the FlexBOSS21 through the EG4® app or EG4 Monitor system for remote system management.

CLOSED-LOOP COMMUNICATIONS

Able to communicate with EG4 48V batteries and other battery brands. A battery firmware update is required for closed-loop communications with LifePower4 batteries.

RAPID SHUTDOWN

The FlexBOSS21 is CSA C22.2#330:2017 and NEC 690.12 ready with its built-in RSD capabilities.

TECHNICAL SPECIFICATIONS

INVERTER	
MODEL	IV-16000-HYB-AW-FX-XX
CEC MODEL #	IV-16000-HYB-AW-FX-XX {240V} IV-16000-HYB-AW-FX-XX {208V}
TYPE	Hybrid
DESIGN TOPOLOGY	High Frequency – Transformerless
AC SYSTEM	
NOMINAL VOLTAGE	120/240 or 208 VAC
FREQUENCY	50/60Hz
PHASE SUPPORT	1ø
AC OUT	
NOMINAL OUTPUT VOLTAGE	120/240 VAC; 120/208 VAC (L1/L2/N required)
MAX. CONTINUOUS OUTPUT	66.7A 16kW
MAX. CONTINUOUS OUTPUT – BATTERY ONLY	50A 6kW
MAX. CONTINUOUS OUTPUT – WITH PV & GRID @25C	66.7A 16kW
PEAK POWER	24000W (.5 sec) 18000W (1 sec) 15000W (6 min) 13200W (12 min)
LOCKED ROTOR AMPS (LRA)	195A
MAX. CONTINUOUS OUTPUT PER-LEG	50A 6kW
POWER FACTOR	.99 @ Full Load
MAX. PASS-THRU CURRENT FROM GRID	90A
REACTIVE POWER ADJUST RANGE	+0.8/-0.8
THD V	<5%
MAX. APPARENT POWER WITH BATTERY	12kVA
MAX. APPARENT POWER WITH PV & GRID	16kVA
AC IN	
NOMINAL GRID VOLTAGE	120/240 VAC 120/208 VAC (L1/L2/N required)
MAX. GRID INPUT POWER TO INVERTER (W/ OUT PASS-THRU)	50A 12kW
MAX. GRID CIRCUIT RATING (W/ PASS-THRU)	90A
MAX. INPUT SHORT CIRCUIT CURRENT RATING	10kA
BATTERY	
NOMINAL VOLTAGE	51.2 VDC
OPERATING VOLTAGE RANGE	40 – 60 VDC
MAX. CHARGE CURRENT (DC AMPS)	250 ADC
MAX. DISCHARGE CURRENT (DC AMPS)	250 ADC
BATTERY CHARGE/DISCHARGE RIPPLE CURRENT	<5%
COMPATIBLE BATTERIES	See www.eq4electronics.com
RECOMMENDED MIN. CAPACITY PER INVERTER	600Ah

EG4 ELECTRONICS

EG4 ELECTRONICS

PV DC IN	
# OF MPPTS	3
RATED CURRENT PER MPPT	26A (MPPT 1) 26A (MPPT 2) 15A (MPPT 3)
INPUTS PER MPPT	3 2 (MPPT1) 2 (MPPT 2) 1 (MPPT 3)
SHORT-CIRCUIT CURRENT RATING PER MPPT	31A (MPPT 1) 31A (MPPT 2) 19A (MPPT 3)
MAX. DC INPUT VOLTAGE*	600 VDC
MPPT DC STARTUP VOLTAGE	200 VDC
MPPT FULL POWER VOLTAGE RANGE**	250 – 440 VDC
MPPT DC OPERATING VOLTAGE RANGE	120 – 240 VDC
MPPT VOLTAGE HIGH PROTECTION***	550 VDC
RECOMMENDED MAX. SOLAR ARRAY POWER (STC)	21kW
RESPONSE	
TRANSFER TIME (GRID TO BATTERY SWITCHING TIME)	20ms (Default), 10ms (Configurable)
OPEN LOOP RESPONSE TIME (OLRT)	<2 sec
TIME TO STEADY STATE	<10 sec
EFFICIENCY	
CEC WEIGHTED EFFICIENCY	97%
MAX. EFFICIENCY: PV TO GRID/LOAD	97%
MAX. EFFICIENCY: BATTERY TO GRID/LOAD	94%
MAX. EFFICIENCY: PV TO BATTERY	94.5%
MAX. EFFICIENCY: AC TO BATTERY	94%
IDLE CONSUMPTION (STANDBY MODE)	<65W @25C
CONTROL & MONITORING	
DISPLAY	Optional EG4 FlexBOSS Screen Kit
USER INTERFACE	App/Web
REMOTE CONNECTIVITY	Wi-Fi, Cellular, and Ethernet (Wi-Fi dongle included by default)
ENVIRONMENTAL	
OPERATING TEMPERATURE RANGE	-13° – 140°F (-25° – 60°C)
STORAGE TEMPERATURE RANGE	-13° – 140°F (-25° – 60°C)
OPERATING HUMIDITY	0 – 95% relative humidity
MAX. ALTITUDE OF OPERATION	<6561 ft. (<2000 m)
ENCLOSURE RATING	NEMA 4X
COOLING METHOD	Fans
NOISE	<50dB @ 3 ft.
PHYSICAL SPECIFICATIONS	
DIMENSIONS (H × W × D)	30.43 × 22.28 × 11.22 in. (773 × 566 × 285 mm)
UNIT WEIGHT	121 lbs. (55 kg)
MOUNTING	Wall mount
MAX # OF INVERTERS IN PARALLEL	16

**Do not exceed the max. DC input voltage specification of 600 VDC. Any damage caused by reaching >600 VDC will not be covered under warranty.*

***When sizing the system, it is best practice to follow the MPPT Full Power Voltage Range specifications, and not the maximum voltage of the MPPT.*

****The value at which the inverter will fault to protect the MPPT from the overvoltage spec of 600 VDC. Ensure **geographical considerations, weather patterns, and panel specifications** are factored into string sizing. See <https://eq4electronics.com/wp-content/themes/hello-elementor/eq4-solar-panel-string-sizer/> for the EG4® string sizing tool*



EG4 ELECTRONICS

WALL MOUNT
ALL WEATHER
LITHIUM BATTERY

The EG4® WallMount All Weather 280Ah batteries are ideal for low-voltage residential outdoor energy storage system (ESS) applications. The batteries use lithium iron phosphate cells with the highest safety performance and an intelligent Battery Management System (BMS) that can monitor and record the voltage of each cell along with the current, voltage, and temperature of the module in real-time. The BMS also contains a passive balance function and an advanced battery control method, both of which improve the performance and longevity of the battery pack.

BUILT-IN
200A BMS

INTEGRATED
600A BUSBARS

82.6MWh
LIFETIME
PRODUCTION*

*10 YEAR
WARRANTY
>8000 CYCLES @
80% DOD

ON-BOARD LCD TOUCH SCREEN

Easy to see BMS monitoring, and selectable closed-loop communications with EG4, Schneider, Sol-Ark, Victron, Growatt, Megarevo, Luxpower, and Deye inverters.

DUAL ON-BOARD FIRE ARRESTORS

Offer fail-safe protection against thermal runaway.

WEATHER-TIGHT QUICK CONNECTS

Included battery cables with outdoor rated connectors allowing for fast, safe, and reliable battery connections.

INTEGRATED SELF-HEATING FEATURE

Heats the battery when the ambient temperature is low. A key feature for outdoor Lithium battery cell operations.

INTEGRATED BUSBARS

The battery design comes manufactured with 600A internal busbars with multiple terminals (4 positive & 4 negative) eliminating the need for external busbars when paralleling batteries and/or multiple inverters.

INNOVATIVE EMERGENCY STOP FUNCTION

The optional ESS disconnect can shut down all batteries and inverters (if equipped with rapid shut down capability) with the press of a button.

THE PERFECT PARTNER TO THE EG4 18kPV

The optional conduit box mates up directly to the connection ports of the inverter allowing a sleek and efficient installation. For other inverters or stand-alone battery installation, the conduit box plugs should be installed.





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VER. 1.1.8 | INFORMATION SUBJECT TO CHANGE WITHOUT NOTICE.
MODEL #: WP-16/280-1AWLL | WP-16/280-1-AWLL

**For information regarding warranty registration on EG4® Electronics products, please navigate to <https://eg4electronics.com/warranty/> and select the corresponding product to begin the registration process.*



EG4 ELECTRONICS

SPECIFICATION SHEET

MODULE OPERATING PARAMETERS			
Parameter	BMS	Recommended Charger Settings	
Total Energy Capacity	14.3kWh @25C, 100% SOC	-	
Voltage	51.2V	-	
Capacity	280Ah ±2%	@25°C ±2°C @ 0.5C	
Charging Voltage (Bulk/Absorb)	56.0V (±0.8V)	56.2V (±0.2V)	
Float	-	54V (±0.2V)	
SOC Cutoff	-	20%*	
Charge Current	140A Max. Continuous	60 – 140A	
Discharge Current	140A Max. Continuous	60 – 140A	

BMS PARAMETERS			
Charge	Spec	Delay	Recovery
Cell Voltage Protection	3.8V	1 sec	3.45V
Module Voltage Protection	60V	1 sec	55.2V
Charge Over-Current 1	>205A	10 sec	-
Charge Over-Current 2	>225A	3 sec	-
Temperature Protection	<23°F or >158°F <-5°C or >70°C	1 sec	>32°F or <140°F >0°C or <60°C

Discharge	Spec	Delay	Recovery
Cell Voltage Protection	2.3V	1 sec	3.1V
Module Voltage Protection	44.8V	1 sec	48V
Discharge Over-Current 1	>205A	10 sec	60 sec
Discharge Over-Current 2	>300A	3 sec	60 sec
Short Circuit	>600A	<0.1 ms	-
Temperature Protection	<-4°F or >167°F <-20°C or >75°C	1 sec	>14°F or <149°F (>-10°C or <65°C)
PCB Temperature Protection	>230°F (>110°C)	1 sec	@ <176°F (<80°C)

GENERAL SPECIFICATIONS			
Parameter	Spec		Condition
Cell Balance	120mA	Passive Balance	Cell Voltage Difference >40mV
Temperature Accuracy	3%	Cycle Measurement	Measure Range: -40°F – ≈212°F (-40°C – ≈100°C)
Voltage Accuracy	0.5%	Cycle Measurement	Cells & Module
Current Accuracy	3%	Cycle Measurement	Measure Range: -200 – 200A
SOC	5%	Integral Calculation	
Power Consumption (Standby)	<300uA	Standby/Storage	
Power Consumption (Operating)	<25mA	Charging/Discharging	
Communication Ports	RS485/CAN	Customizable	

*EG4 recommends this value be set no lower than 20% to maintain the recommended 80% depth of discharge.

EG4 ELECTRONICS

BATTERY HEATER SPECIFICATIONS

Parameter	Spec	Condition
Voltage	56V	-
Power Consumption	224W	-
Internal Battery Temperature	≤32°F (0°C) or ≥41°F (5°C)	Heat On/Heat Off

ENVIRONMENTAL PARAMETERS

Charging Range	32°F to ≈122°F (0°C to ≈50°C)
Discharging Range	-4°F to ≈131°F (-20°C to ≈55°C)
Storage Range	32°F to ≈113°F (0°C to ≈45°C)
Operating Altitude	<16404 ft. (<5000 m)
Ingress Protection	IP65

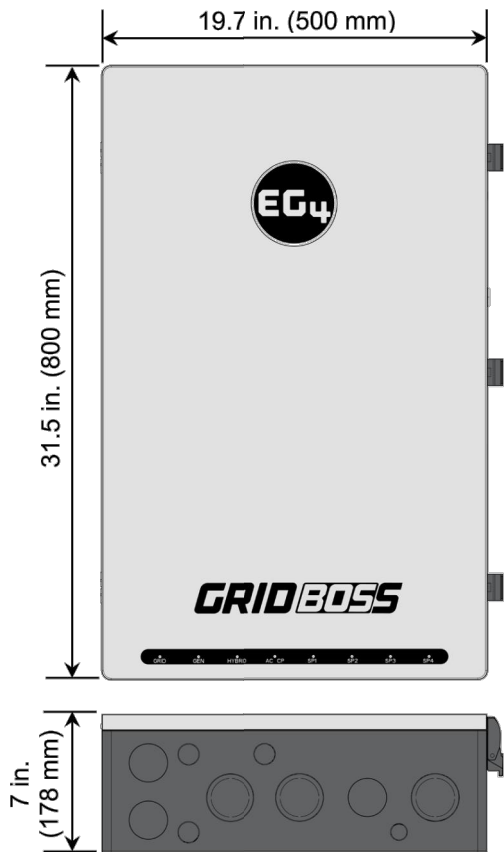
PHYSICAL SPECIFICATIONS

Dimensions (H × W × D)	36.2 × 20.2 × 9.1 in. (92 × 51.2 × 23.2 cm)
Weight	308.6 lbs. (140 kg)
Design Life	>15 years
Cycle Life	>8000 Cycles @ 0.5C 80% DOD
Lifetime Production	82.6MWh**

SAFETY CERTIFICATIONS

Certifications	UL1973, UL9540A (Passed)
----------------	--------------------------

** $(51.2V \times 280Ah / 1000 \times 80\% \times 8000 \text{ cycles} / 1000) 90\% = MWh$



EG4® GRID BOSS

MICRO-GRID INTERCONNECTION DEVICE (MID)

The EG4 GridBOSS Micro-Grid Interconnection Device (MID) simplifies Energy Storage Systems (ESS) by consolidating multiple components into a single, innovative unit. It replaces traditional elements such as the point of common connection, back-fed breakers, feeder taps, tap breakers, supply-side taps, transfer switches, and dedicated combiner panels for grid-in, grid-out, and generator input. As a versatile solution, the GridBOSS serves as the service entrance equipment* when paired with the utility meter, providing a single point of connection for utilities, hybrid inverters, generators, smart loads, and AC-coupled inverters.

200A SERVICE
ENTRANCE*

4 CONFIGURABLE
SMART PORTS

INTEGRATED
GENERATOR SUPPORT

CENTRALIZED ESS CONTROL

Provides a single point of connection for utility, hybrid inverters, generators, smart loads, and AC-coupled inverters.

REDUCED ESS COMPLEXITY

Replaces up to 10 components with one unit, including point of common connection, back-fed breakers, feeder taps, feeder tap breakers, supply side taps & breakers, transfer switches, and dedicated combiner panels for grid-in, load/EPs, and generator input.

SERVICE ENTRANCE RATED

200 Amp service entrance with a 22 kAIC main breaker, acts as service entrance equipment in conjunction with a utility meter and a 200A Eaton breaker CSR2200N (CSR25K).

REMOTE MONITORING

Enable remote monitoring, configuration, and firmware updates through the EG4 mobile app or online monitoring system.

SMART PORTS

Includes load shedding, which disconnects loads during low battery voltage and reconnects on high voltage. Power shedding connects loads when at full SOC and PV flow and disconnects on low SOC or PV loss.



*When used with an Eaton 200A main breaker (model CSR2200N), US only.

TECHNICAL SPECIFICATIONS

GRID	
NOMINAL AC VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT	200A
SERVICE ENTRANCE RATED	22kAIC with 200A Eaton CSR2200N (CSR25K) breaker*
GENERATOR	
NOMINAL VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT	125A
NON-BACKUP	
NOMINAL VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT	200A
BACKUP	
NOMINAL VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT	200A
HYBRID	
NUMBER OF PORTS	3
NOMINAL VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT PER PORT	90A**
SUPPORTED INVERTERS	EG4® 12kPV, 18kPV, FlexBOSS18, FlexBOSS21***
SMART PORTS	
NUMBER OF PORTS	4
NOMINAL VOLTAGE	120/240 VAC (L1/L2/N required)
FREQUENCY	60 Hz
MAXIMUM CURRENT PER PORT	1: 125A 2: 80A 3: 60A 4: 60A
GENERAL DATA	
COMMUNICATION INTERFACE	RS485/Wi-Fi/CAN
IDLE CONSUMPTION	~55W
TRANSFER TIME	~25 ms
INTERNAL BUS RATING	350A (software limited to 200A)
INTERNAL FUSE RATING	315A
OPERATING ALTITUDE	<6561 ft (<2000 m)
RELATIVE HUMIDITY	0 – 100%
OUTDOOR RATING	NEMA 3R
OPERATING AMBIENT TEMPERATURE RANGE	-40°F – 113°F (-40°C – 45°C)
PRODUCT DIMENSIONS (H×W×D)	31.5×19.7×7 in (800×500×178 mm)
UNIT WEIGHT	55 lbs. (25 kg)
STANDARD WARRANTY	10-year standard warranty****

*US only.

**Install a properly sized breaker for the attached inverter: 50A - 12kPV | 70A - 18kPV, FlexBOSS18 | 90A -FlexBOSS21.

***Third-party inverters are not supported and cannot be connected to the hybrid ports.

****For information regarding warranty registration on EG4® Electronics products, please navigate to <https://eg4electronics.com/warranty/> and select the corresponding product to begin the registration process.



The concept of combining photovoltaic arrays with standing seam metal roofing is growing—and for good reasons. A standing seam metal roof has a life expectancy consistent with that of framed PV modules—a 30-year power source on a 40-year roof, along with zero-penetration technology, creates the most sustainable roof system available with alternative power generation, all without compromising the roof manufacturer's warranty! PVKIT® 2.0 is also a great solution for attaching PV modules directly to many exposed fastener roofs when paired with S-5! brackets.

S-5! has introduced a new and improved PVKIT, boasting an improved installation experience for PV mounting technology. The kit comes preassembled with either the MidGrab or EdgeGrab for easier and more efficient installation. The kits were designed with thread lock on the standoff bolt so that the grab will seat to the PV Module frame by using one tool to drive the top bolt, eliminating a step required in the previous PVKIT. The PVKIT 2.0 features bonding teeth, which are more aggressive to secure a better ground path. No lugs or wire required except to connect one string of modules to another and to ground the system.

The S-5 PVKIT 2.0 is built to save you time and money —
The Right Way® to install solar to your metal roof.

PVKIT 2.0 Features:

Pre-assembled kit saves time and money

Only **one tool** needed for installation

Bolt head uses standard hex bit tip which is provided

Improved single piece EdgeGrab installs with ease

Low profile bolt head provides a sleek and clean finish

Also available in black by special order only

MidGrab leaves 1" gap between modules, allowing reduction per ASCE7

UL 2703 Listed

PVKIT® 2.0 MidGrab or EdgeGrab

888-825-3432 | www.S-5.com |

PVKIT 2.0 EdgeGrab Assembly



PVKIT® 2.0: New Design



The PVKIT 2.0 is furnished with the hardware shown at right, excluding the attachment clamp, which is supplied separately. The PVKIT 2.0 is compatible with most common metal roofing materials, including copper.

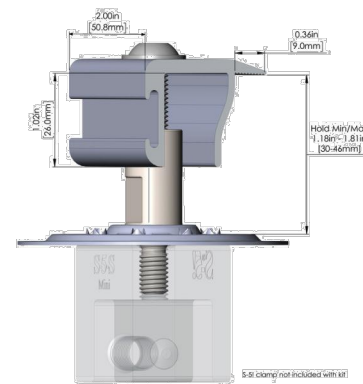
The Module Placement Bevel Guide makes the module placement easier. The mounting disk is multi-directional and rails are not required. The PV grab ears, holding the solar panels in place, are broader to allow for ease of installation and precise module engagement.

Accommodating module thicknesses between 30 and 46mm, the PVKIT 2.0 fits the majority of solar panels on the market. Using the S-5! mini clamps, it fits most standing seam metal roofs. When paired with other S-5! products, the PVKIT 2.0 and EdgeGrab or MidGrab will also work on most exposed fastener including corrugated metal roofs. The MidGrab is designed to fit mid conditions (two adjacent panels), while the new EdgeGrab is designed specifically for end conditions.

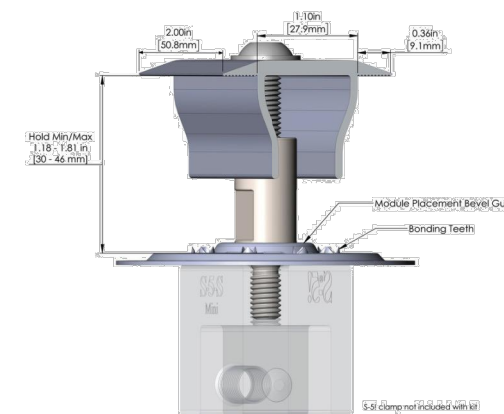
Wind dynamics are complex; thus, each system should be reviewed by a qualified licensed professional who understands wind effects prior to purchase and installation. For more detailed information including specifications, installation instructions, and CAD drawings, visit www.S-5.com or your PVKIT 2.0 distributor.

The PVKIT 2.0 continues to be the easiest, most cost-effective way to install solar panels directly to standing seam and exposed fastener metal roofs, remaining the most popular choice worldwide.

PVKIT 2.0 EdgeGrab



PVKIT 2.0 MidGrab



Certain components featured in illustration may not be UL listed. Due to the variety of attachment needs, S-5-PVKIT 2.0 are sold separately from S-5! clamps.

S-5!® Warning! Please use this product responsibly!

The independent lab test data found at www.S-5.com can be used for load-critical designs and applications.

Products are protected by multiple U.S. and foreign patents. For published data regarding holding strength, fastener torque, patents, and trademarks, visit the S-5! website at www.S-5.com. Copyright 2021, Metal Roof Innovations, Ltd. S-5! products are patent protected.

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The right way to attach almost anything to metal roofs!

S-5![®]

The Right Way

S-5-N Clamp

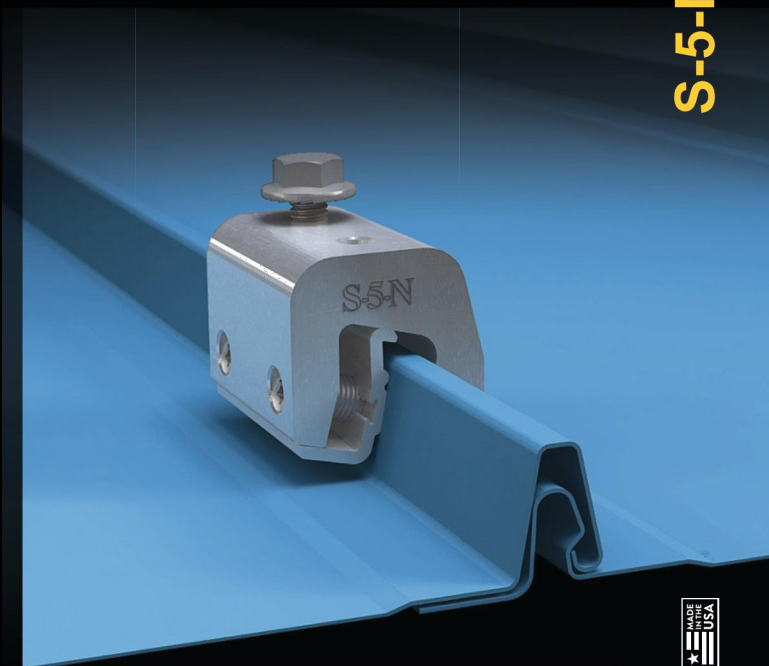
S-5! introduces the new and improved S-5-N clamps. The new design features an innovative insert that ensures a superior fit for new and wider nail strip profiles as well as older ones.

The S-5-N (standard) clamp is the best choice for snow retention and other heavy and load-critical applications. It is designed for use on the most popular 1" nail strip metal roofs, including: Taylor Metal's Easy Lock™, ASC Building Products' Skyline Roofing®, New Tech Machinery's FF100, Schleich 1" Nail Strip, and roofing types with similar profiles.

S-5-N Mini Clamp

The S-5-N Mini offers correct fit to the same profiles as the standard S-5-N but is shorter and has one setscrew rather than two. The Mini is ideal for attaching various rooftop accessories, such as solar arrays, signs, walkways, satellite dishes, lightning protection systems, antennas, rooftop lighting, conduit, condensate lines and other lighter load applications.*

*S-5! Mini clamps are not compatible with, and should not be used with S-5! SnoRail™/SnoFence™ or ColorGard® snow retention systems.



FEATURES AND BENEFITS

- Angled setscrews for easier installation - no special tools required
- Fits seam profiles having base of rib dimension < .82"
- Structural aluminum (6061-T6) clamp body and 300 series SS fasteners offer superior corrosion resistance & strength
- Clamp insert to facilitate installation and fit
- New design ensures straighter clamp position on seam

S-5-N and S-5-N Mini



888-825-3432 | www.S-5.com

S-5![®]

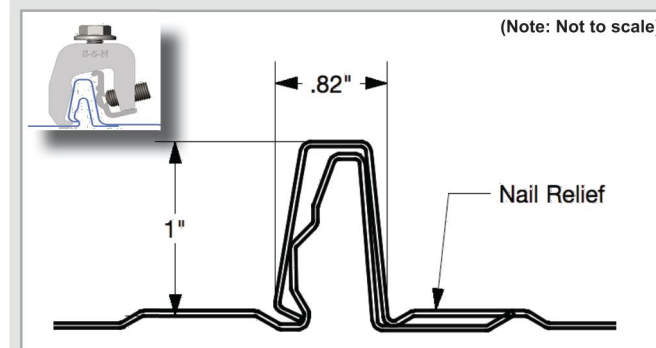
The Right Way

The new and improved S-5-N features angled setscrews, a wider throat, and an insert for easy installation and best fit for wider nail strip profiles.

The **S-5-N** and **S-5-N Mini** clamps are each supplied with the hardware shown to the right. Each box also includes a bit tip for tightening setscrews using an electric screw gun. The S-5-N is a structural aluminum attachment clamp, compatible with most common metal roofing materials (excluding copper). All included hardware is 300 series stainless steel. Please visit www.S-5.com for more information including CAD details, metallurgical compatibilities, and specifications.

S-5![®] holding strength is unmatched in the industry.

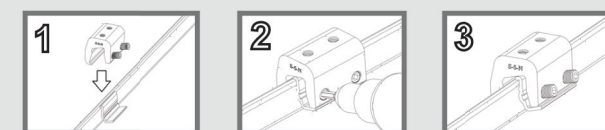
Fits Panels Up to .82"



(NOTE: Seams that exceed maximum allowance at the widest part of the seam will require hand crimping to allow the clamp to fit).

Ease of Installation...in 1...2...3

Installation Simplified: The S-5-N is still just as quick and easy to install as other S-5 clamps. But now, we've angled the setscrews toward the installer, allowing easier access with a screw gun, simplifying tool removal once the setscrew has been tightened - ensuring the clamp sits straighter on the seam. Choose the S-5-N for a non-penetrating solution that protects the roof while providing excellent holding strength.

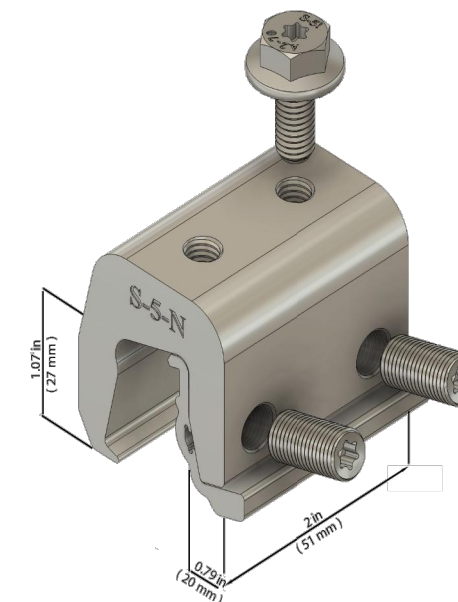


S-5![®] Warning! Please use this product responsibly!

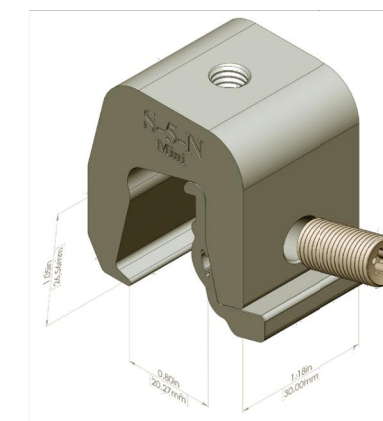
Products are protected by multiple U.S. and foreign patents. Visit the website at www.S-5.com for complete information on patents and trademarks. For maximum holding strength, setscrews should be tensioned and re-tensioned as the seam material compresses. Clamp setscrew tension should be verified using a calibrated torque wrench between 160 and 180 inch pounds when used on 22ga steel, and between 130 and 150 inch pounds for all other metals and thinner gauges of steel. Consult the S-5! website at www.S-5.com for published data regarding holding strength.

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S-5-N Clamp



S-5-N Mini Clamp



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